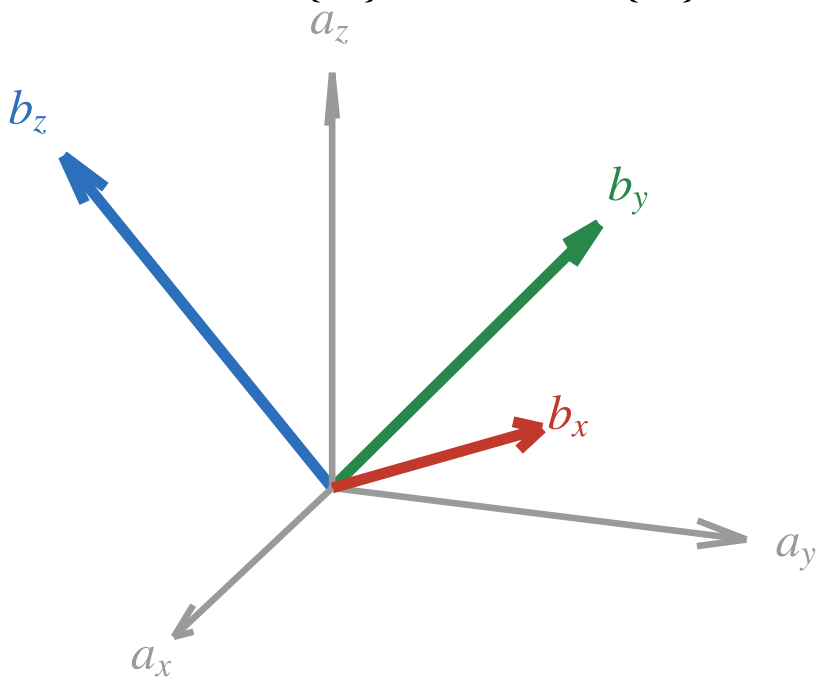
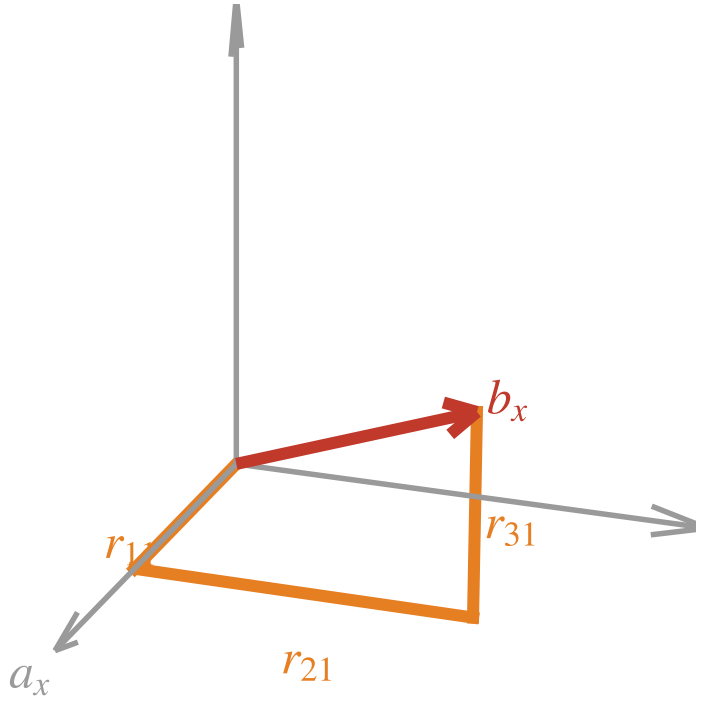
(a) orientation of $\{B\}$ relative to $\{A\}$



(b) column 1 of $R_{AB} = b_x$ in $\{A\}$-coordinates



$$b_x = r_{11}a_x + r_{21}a_y + r_{31}a_z$$

$\Rightarrow$ column 1 of $R_{AB} = b_x$ written in $\{A\}$